

ACCEPTING AI

Come to grips with the future...

SAHIL DAWKA
feedback@digit.in

There's a lot of fear-mongering about Artificial Intelligence and the very future of the human race. Heck, there will be some of it in the articles that follow, and for all we know, they might eventually prove to be well-founded fears. However, this article looks at the flip side, and plays devil's advocate. Why? Because that's how you arrive at truths, by looking at all the evidence, especially the evidence that goes against the premise that "AI is bad for humanity".

JUST A TOOL

Maybe a machine consciousness will someday take offense at this paragraph, but AI is still essentially a productivity tool. It is merely lines of code, written out by humans to enhance the abilities of machines to excel in areas that humans don't. When we think AI we usually think of Skynet, the evil intelligence of the Terminator series, or else the AI of the Matrix world. Even if you think of AI as the characters from the movies A.I. or Bicentennial Man, you're essentially feeding your own narcissism by assuming machines want to be human, when there's no proof that machines even "want" anything.

AI today is so much more than the limited human-killer of fiction, or a poor little machine trying to be a human. AI is used for machine learning, perception, natural language processing, planning, social intelligence, manufacturing, etc. AI is such a broad term, that there's no one discipline that covers it. It is each of the aforementioned fields of study, and all of them as well. AI, as we know it today, is essentially trying to replicate the functions of an intelligence – us.

AI is broadly divided into three categories – Narrow AI (specialized applications of the sub-fields mentioned earlier), General AI (a universal AI with the capabilities of multiple Narrow AIs), and Artificial Superintelligence (think Skynet). Today most AI are of the first category. Virtual Assistants like Siri and Cortana are slowly graduating from Narrow NLP and aiming to be General AI, though they are a long, long way off from even a 5 year old child's intelligence.

WHAT IS "INTELLIGENCE"?

How should we define intelligence itself? It stands to reason that understanding something complex requires an intelligence more complex than the thing that is being understood. This poses a problem. How can we ever understand our brains using our own brains? If you need something more

complex and "higher" than the human brain to understand it, how can we ever hope to create an AI as complex as us? Richard Feynman said, "What I cannot create, I don't understand", so how can we ever create what we will never understand? This itself suggests that unless we can create an AI that learns and enhances itself (evolves), we can never hope to create something more complex than us.

You cannot see your eyes with your own eyes, but you can use a mirror – you will see the general shape of your eyes and how they look and behave, but you will never see the inner workings of your own eyes. This is how we approach AI. We look at external manifestations of our intelligence, try and reverse engineer it using mathematics and logic, and then figure out how to translate that to code. However, it can't just be intelligence in a computer. One school of thought (called the embodied mind thesis) feels that our functional aspects such as movement, visualization, etc., inherently contribute to what we eventually call our 'mind'. The experimental evidence in Daniel Dennett's book *Consciousness Explained* also seems to agree with this idea. Eventually, in order to produce true AI, we might very well need to create the kind of humanoid robot that has been depicted in movies such as *I, Robot* and *Bicentennial Man*.